DAHNet: A Decomposition and Aggregation Hybrid Network for Hyperspectral Object Detection

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***Abstract*—Hyperspectral object detection is important in hyperspectral image analysis. However, it remains challenging because of spectral redundancy and feature coupling. Current studies are limited and lack systematic and general detection frameworks. To address this issue, we propose a hybrid spectral–spatial network named the decomposition and aggregation hybrid network (DAHNet), designed to improve the accuracy and efficiency of hyperspectral object detection (HOD). DAHNet begins with the spectral–spatial compression and enhancement (SSCE) module, which combines depthwise separable and 3-D convolutions to preserve spectral information while capturing key spatial structures, ensuring compact yet discriminative representations. This is followed by feature refinement through the 3-D spectral bottleneck enhancement (3DSBE) module and the scalable spatial bottleneck refinement (SSBR) module, which progressively extract complementary spectral and spatial cues. To effectively integrate the refined features, the cross-domain synergistic fusion (CDSF) module employs an attention mechanism to align spectral and spatial representations, enhancing feature complementarity and semantic consistency. To validate the proposed approach, we constructed an aerial hyperspectral dataset, *SZU_CAR*, and conducted extensive experiments on two additional public datasets. The experimental results demonstrate that DAHNet consistently outperforms existing methods, achieving mean average precisions (mAPs) of 61.5%, 60.5%, and 27.2% on the *SZU_CAR*, HOD3K, and M2SODAI datasets, respectively, with consistent performance gains over representative state-of-the-art approaches. The source code is available at http://szu-hsilab.com/**

***Index Terms*—Attention mechanisms, deep learning, hyperspectral images, object detection.**

I. Introduction

OBJECT detection technology has achieved remarkable progress in RGB imagery, driven by deep learning, leading to high accuracy and efficiency [1], [2], [3]. Representative algorithms such as you only look once (YOLO) and single-shot multibox detector (SSD) have been successfully applied in various fields, including autonomous driving [4], video surveillance [5], and smart homes [6], [7]. These approaches rely on rich RGB information and deep neural networks to learn discriminative features for accurate object recognition and localization. However, RGB images are easily affected by illumination changes, occlusions, and background interference, especially when objects and surroundings share similar spectral characteristics. Such factors often degrade detection performance in real-world applications. In contrast, hyperspectral imaging (HSI) acquires hundreds of narrow spectral bands that record the material-specific reflection properties of each pixel. This rich spectral representation allows HSI to distinguish visually similar objects and maintain robustness against illumination variations and partial occlusions.

Earlier studies on hyperspectral imagery mainly focused on pixel-level target detection (HTD) [8], [9], which identifies whether a target exists in a scene. However, these approaches cannot determine the category or precise location of the target. To overcome this limitation, research has gradually shifted toward hyperspectral object detection (HOD), which extends HTD from pixel-level classification to object-level recognition and localization. By jointly utilizing spectral and spatial information, HOD provides more detailed and reliable detection results. Yan et al. [10] proposed a convolutional neural network (CNN)-based HOD method and established a hyperspectral dataset containing over 400 high-quality images. Their experiments validated the method's effectiveness and demonstrated its potential for industrial and civilian applications. He et al. [11] subsequently introduced spectral-spatial aggregation detection network (S2ADet), which improved HOD performance by jointly aggregating spectral and spatial features. Jang et al. [12] constructed the M2SODAI dataset, which includes synchronized RGB and hyperspectral imagery, and developed a multimodal feature pyramid network (DoubleFPN) for joint feature extraction. Their experiments showed that combining RGB and hyperspectral information significantly enhances detection accuracy, particularly under maritime conditions. Recently, Li et al. [13] proposed SpecDETR, a Transformer-based framework for hyperspectral point object detection. This method treats each pixel's spectral vector as a token and uses a multilayer Transformer encoder to capture spectral–spatial dependencies, achieving substantial performance gains. Li et al. [14] also developed a spatial–spectral fusion algorithm with a visual mamba backbone. Although this method integrates edge-preserving dimensionality reduction and spatial

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feature enhancement techniques, its detection performance under background interference remains limited. Despite these advances, challenges persist in fully exploiting spectral information and improving generalization to mixed background conditions. In addition (ADD) most available datasets contain relatively simple scenes, which restricts the comprehensive validation of algorithm robustness in real-world scenarios.

To enhance the performance and applicability of HOD, we propose a novel framework named the decomposition and aggregation hybrid network (DAHNet). Existing approaches either compress spectral data through dimensionality reduction [11] or jointly process spectral and spatial information without explicit separation [12]. In contrast, DAHNet explicitly decomposes these two domains and introduces a progressive interaction mechanism that enables efficient spectral–spatial collaboration. The network begins with the spectral–spatial compression and enhancement (SSCE) module. This module combines depthwise separable and 3-D convolutions to preserve spectral characteristics while capturing essential spatial features, providing a compact and informative representation for subsequent processing. In the dual-branch refinement stage, DAHNet employs the 3-D spectral bottleneck enhancement (3DSBE) and scalable spatial bottleneck refinement (SSBR) modules. The 3DSBE module focuses on reinforcing spectral features, whereas the SSBR module extracts spatial structures, ensuring comprehensive and accurate feature representation for detection tasks. For feature fusion, the cross-domain synergistic fusion (CDSF) module is introduced to integrate spectral and spatial cues through adaptive attention. It captures dependencies across dimensions and dynamically adjusts feature weights, enhancing target representation and recognition accuracy. The fused features are then fed into the detection head for precise object localization and classification. To validate DAHNet, we constructed the *SZU_CAR* dataset, an aerial hyperspectral dataset designed for vehicle detection. Unlike existing datasets, *SZU_CAR* offers a more challenging configuration for rigorous evaluation, as shown in Fig. 1. The experimental results demonstrate that DAHNet achieves superior detection accuracy compared with existing methods. The main contributions of this work are summarized as follows.

1) We propose a novel hyperspectral detection framework named DAHNet, which employs the CDSF mechanism to achieve efficient spectral–spatial feature extraction and integration. This design substantially improves the representation capability of hyperspectral detectors.
2) We design a dual-branch feature refinement (DBFR) mechanism composed of the 3DSBE and SSBR modules. These modules independently refine spectral and spatial information, ensuring comprehensive and accurate feature representation.
3) We construct a new aerial hyperspectral dataset, *SZU_CAR*, with fine-grained vehicle annotations (Fig. 1). The experimental results show that DAHNet improves average precision (AP) by 5.9% on *SZU_CAR* compared with the baseline model, and achieves additional gains of 11.2% on HOD3K and 14% on M2SODAI. These results demonstrate the superior generalization ability of the proposed framework.

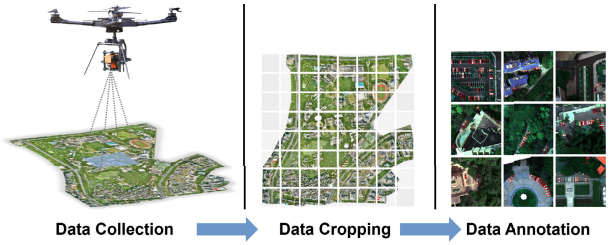


Fig. 1. Data collection process for *SZU_CAR* dataset. It illustrates the various stages involved in capturing and processing hyperspectral images for vehicle detection under challenging background conditions.

II. Related Work

A. RGB-Based Object Detection

Object detection in RGB images [15], [16], [17] has achieved remarkable progress in recent years. This success is largely attributed to the development of deep CNNs [18] and the availability of large-scale annotated datasets [19], [20], [21]. Traditional methods relied on handcrafted features and exhaustive region search, which were computationally expensive and less adaptable to complex scenes. Modern deep-learning-based detectors can be grouped into two main categories, namely, single-stage and two-stage approaches, each offering distinct advantages. Single-stage methods formulate object detection as a direct regression task that predicts bounding-box coordinates and class probabilities simultaneously. Redmon et al. [22] introduced YOLO, which reframed detection as a unified regression problem across the entire image. While YOLO excels in real-time performance, it often struggles with small or overlapping objects. Liu et al. [23] proposed SSD, which detects objects at multiple scales by leveraging different layers of a CNN, effectively balancing speed and accuracy. Feng et al. [24] developed task-aligned one-stage object detection (TOOD) to better align classification and localization, thereby improving detection quality. Chen et al. [25] proposed you only look one-level feature (YOLOF), simplifying the detection pipeline to a single-scale feature map for faster inference while maintaining competitive accuracy.

Two-stage detectors, in contrast, separate the detection process into region proposal and bounding-box refinement. Girshick et al. [26] pioneered this idea with R-CNN, which first generates candidate regions and then classifies each region using a CNN. Although accurate, R-CNN was computationally expensive. Ren et al. [27] addressed this limitation in faster R-CNN by introducing the region proposal network (RPN) for more efficient candidate generation. He et al. [28] extended this framework in Mask R-CNN by adding an instance segmentation branch for pixel-level predictions. Cai and Vasconcelos [29] proposed Cascade R-CNN, which employs multiple sequential detectors to iteratively refine bounding boxes. Sun et al. [30] introduced Sparse R-CNN, using sparse convolutions and attention to reduce computational cost by focusing on limited object queries. Zhang et al. [31] explored query-based detection with dense distinct queries (DDQ), achieving a tradeoff between speed and accu-

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racy in dense scenes. Most recently, Lyu et al. [32] presented real-time multi-task detection (RTMDet), which combines a streamlined backbone and optimized loss design to achieve real-time detection with high accuracy.

B. Transformer-Based Object Detection

Transformer-based object detection methods [33], [34] have become mainstream, with the detection Transformer (DETR) framework being particularly influential. This paradigm combines the advantages of Transformers and CNNs, pioneering a new direction in object detection. Carion et al. [35] introduced DETR as a framework that formulates object detection as a set prediction problem. By applying a Transformer encoder–decoder architecture to the feature maps extracted by a CNN, DETR removes the need for region proposals and anchor boxes. Although DETR is elegant and end-to-end, its initial version required long training and performed poorly on small objects. To improve DETR's efficiency and accuracy, many subsequent variants have been proposed. Zhu et al. [36] presented deformable DETR, which incorporates deformable attention to focus on a sparse set of relevant positions in the feature map. This design reduces computational overhead and enhances the model's ability to capture fine-grained details. Wang et al. [37] proposed anchor DETR, which integrates anchor boxes to better handle objects of varying scales. Li et al. [38] developed denoising DETR, where a query-denoising strategy shortens training time by mitigating the effects of noisy or ambiguous queries. Liu et al. [39] refined this idea by introducing dynamic anchor box DETR (DAB-DETR), which dynamically adjusts anchor boxes during training to improve scale adaptability. Zhang et al. [40] proposed DETR with Improved denoising anchor boxes (DINO), which optimizes the noise-handling strategy of DETR and achieves faster convergence and higher accuracy in cluttered scenes. Chen et al. [41] further introduced group DETR, which groups queries and applies attention within each group to reduce computational complexity and improve scalability.

C. Object Detection Under Complex Backgrounds

Complex backgrounds are often characterized by mixed land-cover types, strong illumination variations, shadows, occlusions, and high spectral or textural similarity between objects and their surroundings. These factors make reliable object detection difficult and frequently lead to false alarms or missed detections. To enhance robustness in such conditions, various strategies have been explored in RGB and multispectral domains. Illumination- and shadow-invariant modeling has been achieved through normalization and adaptive enhancement techniques. These approaches help mitigate brightness inconsistencies in aerial or outdoor scenes [42], [43]. Multiscale and pyramid-based frameworks [29], [44], [45] improve the recognition of small or partially occluded objects by combining fine- and coarse-grained features. Attention mechanisms [31], [46], [47] and context modeling methods further strengthen salient object representation while suppressing background interference. Recent Transformer-based models [35], [40] capture long-range dependencies, enhancing detection stability in heterogeneous and cluttered scenes. In ADD adversarial learning and domain adaptation frameworks [48], [49] improve model generalization across environmental and sensor variations. In HSI, several studies have extended these ideas by exploiting rich spectral–spatial information to better distinguish targets from complex backgrounds. Methods such as S2ADet [11], DoubleFPN [12], and SpecDETR [13] combine spectral cues with multiscale spatial representations to improve discrimination in mixed scenes. However, most existing approaches optimize spectral and spatial features independently, lacking an effective mechanism for collaborative refinement. To overcome this limitation, the proposed DAHNet adopts a decomposition-and-aggregation hybrid architecture that explicitly models spectral–spatial interactions across multiple scales. This design leads to more robust and reliable detection performance in real-world scenarios.

III. Methodology

The overall architecture of DAHNet is illustrated in Fig. 2. This overall design follows a decomposition–aggregation paradigm, which reflects the core idea of DAHNet. In the decomposition stage, hyperspectral features are explicitly separated into spectral and spatial domains through the SSCE module and the dual-branch refinement modules 3DSBE and SSBR, allowing each branch to focus on domain-specific information. In the aggregation stage, the CDSF module adaptively integrates the refined spectral and spatial features via multiscale attention, establishing complementary spectral–spatial correlations. This progressive decomposition–aggregation process enables DAHNet to disentangle and fuse heterogeneous hyperspectral cues more effectively, leading to improved feature representation and detection performance. For multiscale feature extraction and final prediction, DAHNet employs a feature pyramid network (FPN) [44] and a one-stage detection head inspired by [22]. These components ensure scale-adaptive feature representation and stable detection performance across different object sizes. For brevity and visual clarity, the FPN and detector head are not illustrated in Fig. 2, as the primary focus of this work lies in the proposed spectral–spatial decomposition and aggregation framework. Nevertheless, these modules are integrated into the implementation to enhance hierarchical context modeling while maintaining computational efficiency.

A. Spectral–Spatial Compression and Enhancement

In hyperspectral remote sensing imagery, background complexity often arises from mixed land covers, illumination variation, and atmospheric scattering. These factors lead to high spectral redundancy and reduced spatial discriminability. To address these challenges, the proposed SSCE module jointly models spectral and spatial information, suppresses redundant bands, and highlights discriminative features, thereby improving the robustness and stability of subsequent detection.

Let the input hyperspectral image be denoted as $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, where H and W represent the spatial dimensions and C is the number of spectral channels. To reduce computational

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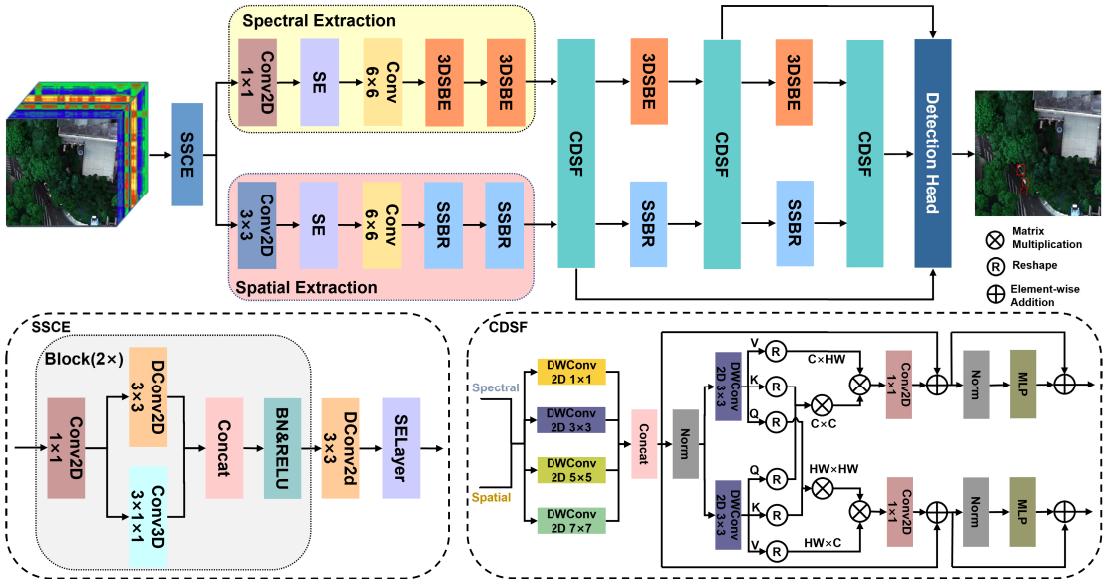


Fig. 2. Overall architecture of the proposed DAHNet. The network includes the SSCE module, responsible for compressing and extracting essential features from hyperspectral data, and the CDSF module, which integrates spectral and spatial features across multiple scales to enhance detection performance under challenging background conditions.

complexity and spectral redundancy, the number of channels is first compressed from C to C' using a 1×1 convolution

$$\mathbf{X}' = \mathrm{Conv}_{1\times1}(\mathbf{X}). \tag{1}$$

The compressed feature map $\mathbf{X}'$ is divided into two branches to extract complementary features. The first branch applies a 3×3 convolution to capture local spatial structures such as edges and textures, while the second branch employs a $3 \times 1 \times 1$ 3-D convolution to model spectral correlations across contiguous bands, enabling the model to distinguish materials that are spatially similar but spectrally distinct. The outputs of the two branches are concatenated along the channel dimension and further enhanced using a depthwise separable convolution

$$\mathbf{F}_{\mathrm{mix}} = \mathrm{Conv}_{3\times3}\left(\mathrm{Concat}\left(\mathrm{Conv}_{3\times3}(\mathbf{X}'), \mathrm{Conv3D}_{3\times1\times1}(\mathbf{X}')\right)\right). \tag{2}$$

To further emphasize important information, a squeeze-and-excitation (SE) channel attention mechanism is introduced. This module first performs global average pooling on $\mathbf{F}_{\mathrm{mix}}$ to obtain a global descriptor vector $\mathbf{z}$ for each channel, and then generates channelwise weight coefficients $\mathbf{s}$ through two fully connected layers

$$\mathbf{z} = \frac{1}{H \times W}\sum_{i=1}^{H}\sum_{j=1}^{W}\mathbf{F}_{\mathrm{mix}}(i, j, :) \tag{3}$$

$$\mathbf{s} = \sigma(\mathbf{W}_{\mathrm{se2}} \cdot \mathrm{ReLU}(\mathbf{W}_{\mathrm{se1}} \cdot \mathbf{z})) \tag{4}$$

where $\mathbf{W}_{\mathrm{se1}}$ and $\mathbf{W}_{\mathrm{se2}}$ denote the weight matrices of the two fully connected layers, and $\sigma(\cdot)$ represents the sigmoid activation function.

Finally, the weight vector $\mathbf{s}$ is applied to each channel of $\mathbf{F}_{\mathrm{mix}}$ through elementwise multiplication to obtain the enhanced feature map

$$\mathbf{F}_{\mathrm{enhanced}} = \mathbf{F}_{\mathrm{mix}} \odot \mathbf{s}. \tag{5}$$

Through this process, the SSCE module adaptively enhances discriminative spectral–spatial representations and suppresses redundant or noisy responses caused by heterogeneous land covers and illumination variations. This design improves the model's ability to extract robust features and enhances detection performance under complex remote sensing conditions.

B. Dual-Branch Feature Refinement

Hyperspectral data contain rich spectral and spatial information, which differ significantly in their representational patterns. Spectral features describe variations across wavelength bands, while spatial features capture local structures and textures. If these heterogeneous cues are processed within a single convolutional pathway, spectral and spatial information can easily become entangled, reducing discriminative clarity. To address this issue, DAHNet introduces a DBFR module, which separates and refines spectral and spatial information through two dedicated branches, the 3DSBE and SSBR, enabling independent learning and complementary enhancement.

The enhanced feature map $\mathbf{F}_{\mathrm{enhanced}}$ from the previous stage is fed into both branches, denoted as $\mathbf{I}_{\mathrm{SPE}}$ (spectral) and $\mathbf{I}_{\mathrm{SPA}}$ (spatial), each with dimensions $H \times W \times C'$. The ith layer outputs of these branches are formulated as

$$\mathbf{F}_{\mathrm{SPE}}^{i} = \phi_{\mathrm{branch_spe}}(\mathbf{I}_{\mathrm{SPE}}; \theta_{\mathrm{SPE}}) \tag{6}$$

$$\mathbf{F}_{\mathrm{SPA}}^{i} = \phi_{\mathrm{branch_spa}}(\mathbf{I}_{\mathrm{SPA}}; \theta_{\mathrm{SPA}}) \tag{7}$$

where $\phi_{\mathrm{branch_spe}}$ and $\phi_{\mathrm{branch_spa}}$ represent the spectral and spatial feature extraction functions with trainable parameters θ_{SPE} and θ_{SPA}, respectively. This decoupled design allows the network to model heterogeneous cues independently, minimizing interference and preserving feature purity for subsequent fusion.

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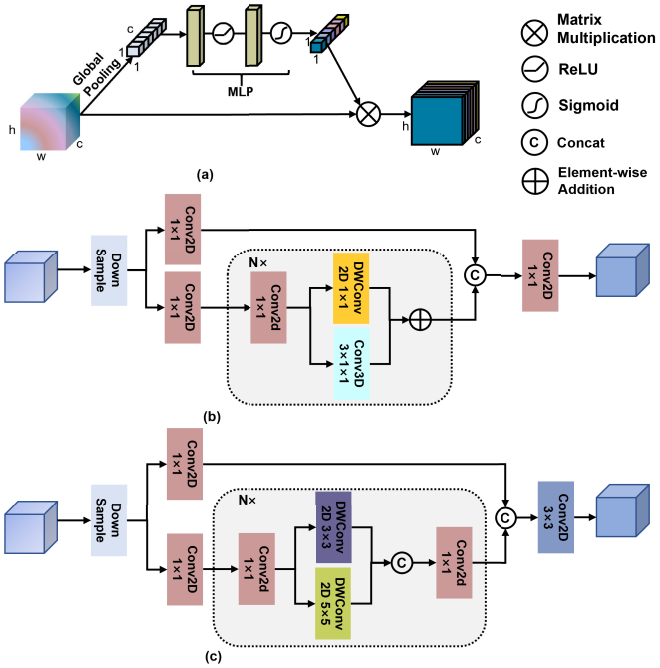


Fig. 3. (a) Architecture of the SE layer module. (b) Architecture of the 3DSBE module. (c) Architecture of the SSBR module.

1) 3-D Spectral Bottleneck Enhancement: The goal of the 3DSBE module is to strengthen cross-band relationships in hyperspectral data. Traditional 2-D convolutions operate purely in the spatial domain, failing to capture spectral dependencies, whereas naive 3-D convolutions incur heavy computational cost and redundant processing. Hence, 3DSBE adopts a hierarchical strategy that first downsamples spatial resolution to reduce redundancy, then refines spectral representations through a bottleneck structure, and finally applies lightweight 3-D convolutions to model interband dependencies efficiently.

As illustrated in Fig. 3(b), the input $\mathbf{F}_{\text{SPE-in}} \in \mathbb{R}^{H\times W\times C'}$ is first downsampled to $H/2 \times W/2$ and expanded to $2C'$ channels. This tradeoff between spatial and channel dimensions enables the model to emphasize bandwise representation while suppressing unnecessary spatial redundancy. The downsampled feature map $\mathbf{Y}$ is processed by two parallel 1×1 convolutional paths: one produces the bottleneck input $\mathbf{Z}$, and the other preserves shallow cues for later fusion, which helps prevent the loss of low-level information during compression.

Within the bottleneck, $\mathbf{Z}$ is first transformed via a 1×1 convolution to yield an intermediate feature $\mathbf{U}$. Then, two parallel operations are applied: a grouped 1×1 convolution captures compact channelwise dependencies, and a $3 \times 1 \times 1$ 3-D convolution models interband correlations along the spectral dimension. Their outputs are merged through elementwise summation

$$\mathbf{S} = \text{DWConv}_{1\times1}(\mathbf{U}) + \text{Conv3D}_{3\times1\times1}(\mathbf{U}). \tag{8}$$

This hybrid operation allows fine-grained local filtering and spectral correlation modeling to coexist. The grouped convolution enhances channel selectivity, while the 3-D convolution provides global spectral coupling, together producing compact yet expressive spectral representations. The refined feature $\mathbf{S}$ is concatenated with the auxiliary branch output and passed through a 1×1 convolution to restore feature compactness

$$\mathbf{F}_{\text{SPE-out}} = \text{Conv}_{1\times1}\left(\text{Concat}\left(\mathbf{S}, \text{Conv}_{1\times1}(\mathbf{Y})\right)\right). \tag{9}$$

Through this stepwise decoupling–aggregation mechanism, 3DSBE effectively strengthens the discriminability of spectral features without increasing computational complexity, providing structured and band-aware representations for the subsequent fusion stage.

2) Scalable Spatial Bottleneck Refinement: The SSBR module aims to enhance spatial representation by capturing both fine-grained details and broad contextual structures. Objects in hyperspectral imagery often exhibit significant variation in size and geometry, and using a single convolution scale may limit the receptive field or miss structural continuity. To mitigate this, SSBR adopts a multiscale bottleneck design that allows simultaneous perception at multiple spatial scales.

As shown in Fig. 3(c), the input feature is first passed through two parallel 1×1 convolutions to reduce channels from $2C'$ to C'. Here, $\mathbf{X}_1$ serves as the main branch for multiscale modeling, while $\mathbf{X}_2$ acts as an auxiliary shortcut preserving structural alignment. This "main–auxiliary" arrangement reduces computational load and provides stable reference features for fusion.

The main branch $\mathbf{X}_1$ is then processed by convolution kernels of sizes 1×1, 3×3, and 5×5. Small kernels focus on local textures, whereas large ones capture broader contextual patterns. The outputs from different receptive fields are concatenated and fused using a 1×1 convolution

$$\mathbf{F}_{\text{SPA-mid}} = \text{Conv}_{1\times1}\left(\text{Concat}\left(\text{Conv}_{3\times3}(\mathbf{X}_1)\ \text{Conv}_{5\times5}(\mathbf{X}_1)\right)\right). \tag{10}$$

By integrating features from multiple scales within the same layer, the network learns spatial cues that are both detailed and structurally coherent, improving the continuity of object boundaries and geometric consistency. Finally, the fused feature $\mathbf{F}_{\text{SPA-mid}}$ is concatenated with $\mathbf{X}_2$ and processed by a 3×3 convolution to produce the refined spatial output

$$\mathbf{F}_{\text{SPA-out}} = \text{Conv}_{3\times3}(\text{Concat}(\mathbf{F}_{\text{SPA-mid}}, \mathbf{X}_2)). \tag{11}$$

This design preserves local textures while maintaining global structural integrity, providing stable and geometry-aware spatial representations for subsequent spectral–spatial fusion.

C. Cross-Domain Synergistic Fusion

In HOD, spectral and spatial features exhibit distinct representational characteristics. Spectral information reflects material composition across wavelength bands, whereas spatial cues describe geometric structures and textural patterns. When these heterogeneous features are simply concatenated or combined through weighted summation, semantic misalignment, and redundant responses often arise, preventing the network from fully exploiting their complementary nature. To address this limitation, the proposed CDSF module employs an

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attention-based mechanism to construct bidirectional dependencies between the spectral and spatial domains, thereby enabling adaptive alignment and context-aware fusion of cross-modal information.

Let the spectral and spatial features from the dual branches be $\mathbf{F}_{\text{SPE-out}}$ and $\mathbf{F}_{\text{SPA-out}}$, each with dimensions $\mathbb{R}^{H/2\times W/2\times 2C'}$. The CDSF first introduces a multiscale depthwise separable convolution unit, which applies parallel convolutional kernels with different receptive fields to capture both local structures and global contexts. Compared with single-scale convolution, this design enhances the diversity of spatial–spectral cues while maintaining structural integrity. The outputs of the parallel branches are concatenated along the channel dimension to form multiscale representations

$$\mathbf{F}_{\text{SPE-ms}} = \text{Concat}\left(\mathbf{F}^{(1)}_{\text{SPE-out}}, \mathbf{F}^{(3)}_{\text{SPE-out}}, \mathbf{F}^{(5)}_{\text{SPE-out}}, \mathbf{F}^{(7)}_{\text{SPE-out}}\right) \quad (12)$$

$$\mathbf{F}_{\text{SPA-ms}} = \text{Concat}\left(\mathbf{F}^{(1)}_{\text{SPA-out}}, \mathbf{F}^{(3)}_{\text{SPA-out}}, \mathbf{F}^{(5)}_{\text{SPA-out}}, \mathbf{F}^{(7)}_{\text{SPA-out}}\right) \quad (13)$$

where Concat$(\cdot)$ denotes the concatenation operation along the channel dimension, and $\mathbf{F}_{\text{SPE-ms}}$ and $\mathbf{F}_{\text{SPA-ms}}$ represent the aggregated multiscale spectral and spatial feature tensors, respectively. The superscripts (1), (3), (5), and (7) indicate that the corresponding features are extracted by depthwise separable convolutions with kernel sizes of 1×1, 3×3, 5×5, and 7×7, respectively.

Subsequently, the features are flattened along spatial dimensions and supplemented with learnable positional embeddings to preserve geometric correspondence

$$\mathbf{T}_{\text{SPE}}, \mathbf{T}_{\text{SPA}} \in \mathbb{R}^{N\times C_t}, \quad N = \frac{H}{2} \times \frac{W}{2} \quad (14)$$

where N is the number of spatial tokens, and C_t is the token embedding dimension projected from the multiscale features. $\mathbf{T}_{\text{SPE}}$ and $\mathbf{T}_{\text{SPA}}$ denote the spectral and spatial token sequences, which serve as inputs to the subsequent cross-domain attention module. Within the CDSF block, a bidirectional cross-attention mechanism is employed to explicitly model interdomain dependencies. In this process, spectral tokens are guided by spatial structures to achieve discriminative focus, while spatial tokens are enhanced under spectral constraints that improve boundary precision and intraclass consistency. In the spectral-to-spatial direction, the spectral tokens $\mathbf{T}_{\text{SPE}}$ serve as queries, and the spatial tokens $\mathbf{T}_{\text{SPA}}$ act as keys and values

$$\mathbf{Q} = \mathbf{T}_{\text{SPE}}\mathbf{W}_Q \quad (15)$$

$$\mathbf{K} = \mathbf{T}_{\text{SPA}}\mathbf{W}_K \quad (16)$$

$$\mathbf{V} = \mathbf{T}_{\text{SPA}}\mathbf{W}_V \quad (17)$$

where $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$ are the learnable projection matrices used to generate the query, key, and value representations, respectively. The attention response for each head i is computed as

$$h_i = \text{Softmax}\left(\frac{\mathbf{Q}^i\left(\mathbf{K}^i\right)^T}{\sqrt{d_K}}\right)\mathbf{V}^i \quad (18)$$

where d_K denotes the dimensionality of the key vectors used for scaling normalization. The responses of all n attention heads are concatenated and projected as

$$\mathbf{T}_{\text{fuse}} = \text{Concat}\left(h_1, h_2, \ldots, h_n\right)\mathbf{W}^o \quad (19)$$

where $\mathbf{W}^o$ is the output projection matrix producing the fused token representation $\mathbf{T}_{\text{fuse}}$.

Through this bidirectional interaction, spectral features are adaptively refined under the guidance of spatial structures, suppressing redundant background responses and emphasizing spectrally distinctive regions. At the same time, spatial features gain discriminative consistency and clearer boundaries under spectral modulation. Compared with conventional one-way fusion or static weighting strategies, this cross-domain design enables dynamic semantic alignment and mutual reinforcement between spectral and spatial representations, yielding a unified and coherent feature embedding.

To further enhance nonlinear feature modeling, a lightweight feed-forward network is applied for tokenwise reconstruction

$$\mathbf{T}_{\text{out}} = \text{FC}_2\left(\text{ReLU}\left(\text{FC}_1\left(\mathbf{T}_{\text{fuse}}\right)\right)\right) \quad (20)$$

and the refined features are reshaped back to their spatial form

$$\mathbf{F}_{\text{fuse}} = \text{Reshape}\left(\mathbf{T}_{\text{out}}\right) \in \mathbb{R}^{\frac{H}{2}\times\frac{W}{2}\times 2C'}. \quad (21)$$

The fused feature map $\mathbf{F}_{\text{fuse}}$ is then forwarded to the detection head for the final prediction stage. By integrating multiscale convolution for contextual aggregation, cross-domain attention for bidirectional interaction, and feed-forward reconstruction for refinement, the CDSF establishes a hierarchical fusion pathway that progressively aligns spectral–spatial semantics and strengthens discriminative hyperspectral representations for accurate object detection.

D. Loss Function

After fusing the spectral and spatial features, the enhanced representations are passed into the detection head for object localization and classification. The overall loss function of DAHNet follows the YOLO paradigm and is defined as:

$$L = \lambda_{\text{box}}L_{\text{box}} + \lambda_{\text{conf}}L_{\text{conf}} + \lambda_{\text{cls}}L_{\text{cls}} \quad (22)$$

where L_{box}, L_{conf}, and L_{cls} denote the bounding box regression loss, confidence loss, and classification loss, respectively. The coefficients λ_{box}, λ_{conf}, and λ_{cls} balance the contribution of each component.

The bounding box regression loss L_{box} is implemented using complete intersection over union (CIoU) loss [50], which jointly considers the overlap area, center distance, and aspect ratio between the predicted box B_p and the ground truth (GT) B_g

$$L_{\text{box}} = 1 - \text{CIoU}\left(B_p, B_g\right). \quad (23)$$

The confidence loss L_{conf} measures the accuracy of objectness prediction and is computed by binary cross-entropy (BCE) as

$$L_{\text{conf}} = -\left[y\log\left(p\right) + \left(1-y\right)\log\left(1-p\right)\right] \quad (24)$$

where y indicates whether an object exists in a given anchor, and p is the predicted confidence score.

The classification loss L_{cls} evaluates the correctness of the predicted category, also using the BCE formulation

$$L_{\text{cls}} = -\sum_{k=1}^{N_{\text{cls}}}\left[y_k\log\left(p_k\right) + \left(1-y_k\right)\log\left(1-p_k\right)\right] \quad (25)$$

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TABLE I
COMPARISON OF HYPERSPECTRAL DATASETS

Dataset	Year	Images	Bands	Wavelength (nm)	Size	Classes	Instances	Scene
HOD3K	2023	3,242	16	470∼620	512×256	3	15,149	Urban
M2SODAI	2023	1,257	127	400∼1000	224×224	2	11,527	Maritime
SZU_CAR	**2024**	**605**	**112**	**400∼1000**	**640×640**	**1**	**2,472**	**Urban**

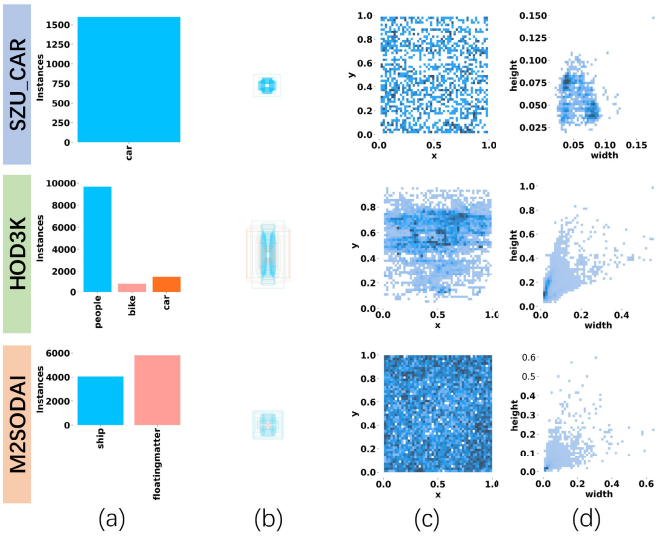


Fig. 4. Associated statistics of three datasets: (first row) *SZU_CAR*, (second row) HOD3K, and (third row) M2SODAI. (a) Number of objects of each type in the training set. (b) Size and number of the real bounding boxes of different categories in the training set. (c) Position of the center point of the object in relation to the whole image. (d) Aspect ratio of objects in an image relative to the whole image.

where N_{cls} is the total number of categories, and y_k and p_k denote the GT and predicted probabilities for class k, respectively.

These loss components collaboratively guide the network optimization process, where L_{box} emphasizes precise localization, L_{conf} enhances the objectness discrimination, and L_{cls} refines the category prediction. By jointly minimizing these terms, DAHNet achieves robust and accurate object detection in hyperspectral imagery.

IV. EXPERIMENTS

A. Datasets

In this study, three HOD datasets are used to evaluate the effectiveness of the proposed method. They include one self-collected dataset, *SZU_CAR*, and two publicly available datasets, HOD3K and M2SODAI. Table I summarizes the main characteristics of these datasets used in our experiments.

SZU_CAR dataset is specifically designed for vehicle object detection in scenarios characterized by high background complexity. The dataset includes hyperspectral images from Yuehai and Canghai campuses of Shenzhen University, captured using a unmanned aerial vehicle (UAV) equipped with a Specim FX10 hyperspectral camera. The images cover 112 continuous spectral bands ranging from 400 to 1000 nm and have a spatial resolution of 0.1 m/pixel with a size of 640×640 pixels. The dataset consists of 605 images, which are manually annotated and split into a training set (417 images) and a testing set (188 images) in a 7:3 ratio. As shown in the first row of Fig. 4, the dataset contains a single object category ("car"), with approximately 1500 instances. The bounding boxes are relatively small and concentrated, with the majority of widths and heights distributed between 0.05 and 0.15 in normalized scale, which limits the available spatial context for target discrimination. The center points of the targets are evenly distributed across the entire image, reflecting diverse object locations in real urban scenes, while the aspect ratios of bounding boxes are predominantly close to 1:1, consistent with the rectangular shape of vehicles in aerial views. Beyond these basic statistics, *SZU_CAR* presents several quantitative challenges arising from real-world urban environments. The background consists of heterogeneous land-cover types, including roads, buildings, vegetation, and parking areas, whose spectral responses partially overlap with those of vehicles, leading to background-induced spectral confusion. Partial occlusions frequently occur due to trees, building shadows, and vehicle overlaps, and illumination conditions vary across scenes because of different acquisition times and viewing angles. These factors jointly increase appearance variability and detection difficulty, making *SZU_CAR* a valuable benchmark for evaluating HOD methods in realistic urban environments. Since the dataset contains a single object category with relatively consistent object sizes and uniformly distributed spatial locations, the primary detection difficulty stems from background-induced spectral confusion rather than scale variation. This property motivates the design of DAHNet, which emphasizes early spectral–spatial compression and subsequent domain-specific refinement to suppress background interference and enhance target discrimination.

HOD3K dataset, developed by China University of Geosciences, is specifically tailored for HOD tasks. It contains 3242 hyperspectral images covering 16 spectral bands with wavelengths ranging from 470 to 620 nm. The dataset features diverse natural scenes, such as urban roads, campuses, and residential areas, with targets classified into three categories: humans, cars, and bicycles, resulting in a total of 15 149 annotated objects. The dataset is split into training, validation, and testing sets with a 70%, 10%, and 20% ratio, respectively. As shown in the second row of Fig. 4, the "people" category accounts for the majority of annotations, with nearly 10 000 instances, while "bike" and "car" are less represented. Bounding box sizes vary significantly across

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categories, with "people" having smaller dimensions compared to the other two categories. The center points of the targets are primarily concentrated in the central regions of the images, reflecting the typical positioning of objects in urban and campus environments. The aspect ratios of bounding boxes demonstrate notable diversity, with "people" exhibiting taller and narrower shapes, while "bike" and "car" have dimensions closer to a square. This distribution reveals pronounced variations in object scales and aspect ratios across categories, which increases the difficulty of spatial structure modeling and motivates the use of multiscale spatial refinement in the SSBR module of DAHNet.

Developed by POSTECH and KRISO in South Korea, the M2SODAI dataset is a multimodal maritime object detection dataset that includes synchronized RGB and hyperspectral imagery. Its primary focus is detecting maritime targets—such as ships and floating objects—within challenging oceanic environments. The original hyperspectral data span 400–1000 nm and comprise 127 spectral bands. For efficiency, a reduced 30-band version is employed after dimensionality reduction. In total, the M2SODAI dataset consists of 1257 pairs of synchronized RGB and hyperspectral images, with 11 527 annotated maritime targets. The split for training, validation, and testing comprises 1007, 125, and 125 pairs of images, respectively. As shown in the third row of Fig. 4, the dataset contains two categories, "ship" and "floating matter," with a balanced number of annotations for each. Bounding box sizes vary widely, with "ship" typically having larger widths, while "floating matter" has smaller, more compact dimensions. The center points of targets are evenly distributed throughout the images, reflecting the random placement of objects in oceanic scenes. The aspect ratios highlight the morphological differences between the categories, with "ship" exhibiting wider shapes and "floating matter" being closer to square proportions. These characteristics make M2SODAI a challenging benchmark for maritime object detection, especially for small targets whose limited spatial extent and low contrast against complex maritime backgrounds pose significant difficulties. Ultimately, hyperspectral images were cropped to $224 \times 224 \times 30$ for computational efficiency. This thorough data preparation pipeline ensures that M2SODAI offers a challenging and realistic setting for multimodal maritime object detection experiments.

B. Experimental Setup

1) Implementation Details: In this study, our experiments were implemented using the PyTorch 1.12.1 framework and conducted on a single NVIDIA A40 GPU. We processed three datasets: SZU_CAR, HOD3K, and M2SODAI, all used for HOD tasks. For the SZU_CAR and HOD3K datasets, the input image size was set to 640×640 pixels, while for the M2SODAI dataset, the input image size was set to 320×320 pixels. The model was trained for a total of 300 epochs with a batch size of 8. We employed the stochastic gradient descent (SGD) [51] optimizer with an initial learning rate of 0.01, a momentum coefficient of 0.937, and a weight decay of 0.0005. During training, the learning rate gradually decreased according to a polynomial learning rate schedule, where the learning rate was reduced by a factor of 0.9 at the end of each training epoch. To enhance the model's generalization ability, we applied Mosaic data augmentation, which increases the diversity of the training data by combining multiple images into one. Additionally, random horizontal flipping, scale transformation, and color jitter were used as data augmentation techniques to further improve the model's robustness across different scenarios. For the loss function, we used BCE loss for handling classification and localization losses, while the confidence loss was managed using the CIoU [52] loss function to improve the precision and stability of object detection.

2) Evaluation Metrics: In this study, we used mean AP (mAP), mAP50, and mAP75 as the primary evaluation metrics to assess the performance of our model. mAP represents the mean AP at multiple IoU thresholds (ranging from 0.50 to 0.95 with a step size of 0.05) and is used to measure the overall performance of the model. mAP50 and mAP75 indicate the mAP at an IoU threshold of 0.50 and 0.75, respectively. These metrics are calculated based on the relationship between precision and recall.

C. Experimental Results

In this section, we compare the proposed DAHNet model with a range of recent object detection methods on multiple hyperspectral datasets. The comparative methods cover both two-stage and one-stage detection paradigms. The two-stage detectors include Faster R-CNN [27], DDQ [31], and DoubleFPN [12], while the one-stage detectors include RTMDet [32], YOLOv5 [53], YOLOF [25], TOOD [24], DINO [40], RTDETR [54], S2ADet [11], YOLOv8 [55], YOLOv11 [56], and SpecDETR [13]. These representative detectors encompass a wide spectrum of modern architectures and detection strategies, providing a solid benchmark for comprehensively evaluating the effectiveness of the proposed model. In ADD, an internal baseline model is established to verify the actual contribution of each proposed module. This model adopts the same dual-branch overall framework as DAHNet, but its backbone is the original cross-stage partial Darknet-53 (CSPD53) architecture. The two branches are fused through elementwise ADD, and the input hyperspectral data are directly fed into both branches without any spectral–spatial compression or enhancement operations. This baseline design serves to isolate and quantitatively evaluate the performance gains introduced by the proposed SSCE, DBFR, and CDSF modules in the overall network architecture.

The results presented in Table II indicate that DAHNet achieves the best overall detection performance on the SZU_CAR dataset. Specifically, DAHNet attains an mAP of 61.5%, surpassing RTMDet by 12.5% and YOLOv5 by 6.1%. Compared with the internal baseline model, which achieves 55.6%, DAHNet improves by 5.9%. The baseline shares the same dual-branch structure as DAHNet but does not include the SSCE, DBFR, and CDSF modules. This comparison directly verifies the contribution of these modules to the overall improvement in detection accuracy. The results demonstrate that the proposed spectral–spatial modeling mechanism effectively enhances feature representation, enabling the network to

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TABLE II
PERFORMANCE COMPARISON ON THE *SZU_CAR* DATASET

Method	Backbone	Type	mAP	mAP_{50}	mAP_{75}	FLOPs (G)↓	FPS (↑)
Faster R-CNN	ResNet50	Two-stage	21.1	44.4	16.6	126.0	32.9
DDQ	ResNet50	Two-stage	45.7	66.0	55.1	175.7	15.4
YOLOv5	CSPD53	One-stage	55.4	80.7	65.4	79.5	18.5
YOLOF	ResNet50	One-stage	7.6	22.0	3.9	**74.2**	**45.2**
TOOD	ResNet50	One-stage	29.0	53.0	28.7	114.0	24.6
RTMDet	CSPNeXt	One-stage	49.0	70.1	60.2	83.2	26.7
DINO	ResNet50	One-stage	36.8	59.8	41.2	154.0	17.1
YOLOv8	C2f-SPPF	One-stage	56.4	82.6	66.4	177.7	21.1
YOLOv11	GELAN	One-stage	57.0	79.6	66.3	99.4	23.4
RTDETR	ResNet50	One-stage	54.0	79.2	64.1	109.9	26.8
SpecDETR	—	One-stage	36.5	58.7	41.5	1059.0	3.2
Baseline	CSPD53	One-stage	55.6	80.7	65.2	139.9	10.1
DAHNet (Ours)	DAHNet	One-stage	**61.5**	**87.0**	**72.3**	150.0	22.5

TABLE III
PERFORMANCE COMPARISON ON THE HOD3K DATASET

Method	Backbone	Type	mAP	mAP_{50}	FLOPs (G)↓	FPS (↑)
Faster R-CNN	ResNet50	Two-stage	41.9	75.7	95.1	33.7
DDQ	ResNet50	Two-stage	43.6	76.8	67.5	13.0
YOLOv5	CSPD53	One-stage	52.3	83.7	56.9	33.4
YOLOF	ResNet50	One-stage	10.9	23.3	**21.9**	80.8
TOOD	ResNet50	One-stage	35.9	67.8	83.1	34.0
RTMDet	CSPNeXt	One-stage	52.8	85.2	80.3	35.6
DINO	ResNet50	One-stage	41.8	71.7	68.4	18.4
YOLOv8	C2f-SPPF	One-stage	52.8	86.2	166.4	**287.3**
YOLOv11	GELAN	One-stage	51.5	84.8	88.1	273.4
RTDETR	ResNet50	One-stage	48.8	80.4	104.2	166.4
S2ADet	S2ANet	One-stage	49.3	82.0	235.1	30.5
SpecDETR	—	One-stage	33.2	54.6	978.0	3.5
Baseline	ResNet50	One-stage	53.3	85.0	94.7	58.5
DAHNet (Ours)	DAHNet	One-stage	**60.5**	**91.2**	149.0	27.4

capture cross-band and multiscale semantic information more comprehensively, which leads to stable and accurate detection in challenging scenarios. At higher IoU thresholds, DAHNet achieves an mAP75 of 72.3%, outperforming YOLOv8 by 5.9%. This improvement highlights the model's advantage in precise object localization, small-object detection, and occlusion handling. Furthermore, DAHNet obtains an mAP50 of 87.0%, exceeding YOLOv5 by 6.3% and YOLOv8 by 4.4%, indicating that the model maintains high stability and recall under lower IoU thresholds. In terms of computational complexity and inference efficiency, DAHNet records 150.0 giga floating-point operations (GFLOPs) and achieves 22.5 frames per second (FPS), exhibiting a favorable balance between accuracy and efficiency. In contrast, SpecDETR, a backbone-free DETR-like hyperspectral detector, incurs extremely high computational cost of 1059.0 GFLOPs and exhibits very low inference speed of 3.2 FPS, while achieving a relatively low mAP of 36.5%, indicating limited efficiency and scalability for instance-level HOD on the *SZU_CAR* dataset. Although YOLOF achieves higher speed at 45.2 FPS with lower computational cost (74.2 GFLOPs), its detection accuracy is only 7.6%, which is substantially lower than that of DAHNet. RTMDet, YOLOv11, and RTDETR achieve mAPs of 49.0%, 57.0%, and 54.0%, respectively, all far below DAHNet. These results confirm that DAHNet significantly improves detection performance while maintaining reasonable computational overhead, demonstrating the effectiveness and stability of the proposed decomposition–aggregation strategy for spectral–spatial feature modeling. The findings also provide a solid foundation for further validation on larger scale datasets such as HOD3K and M2SODAI.

Table III presents the detection results of different methods on the HOD3K dataset. DAHNet achieves an mAP of 60.5%, which is 7.2% higher than the internal baseline and 11.2%

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TABLE IV
PERFORMANCE COMPARISON ON THE M2SODAI DATASET

Method	Backbone	Type	mAP	mAP_{50}	AP_S	AP_M	FLOPs (G)↓	FPS (↑)
Faster R-CNN	ResNet50	Two-stage	4.8	14.0	5.0	2.9	35.3	60.4
DDQ	ResNet50	Two-stage	7.9	19.8	7.7	22.6	32.1	23.8
DoubleFPN	ResNet50	Two-stage	13.2	30.6	12.3	29.9	150.1	30.1
YOLOv5	CSPD53	One-stage	23.5	59.0	23.3	33.1	**15.1**	82.4
TOOD	ResNet50	One-stage	8.1	23.9	8.3	6.0	21.9	37.7
RTMDet	CSPNeXt	One-stage	20.8	48.0	20.8	41.4	20.2	36.6
DINO	ResNet50	One-stage	5.4	16.9	5.7	2.4	41.6	32.6
YOLOv8	C2f-SPPF	One-stage	25.7	56.3	20.0	**45.9**	168.0	**143.7**
YOLOv11	GELAN	One-stage	24.9	54.9	19.9	43.2	89.8	137.4
Baseline	ResNet50	One-stage	23.4	56.6	23.1	33.2	25.3	56.8
DAHNet (Ours)	DAHNet	One-stage	**27.2**	**64.0**	**26.7**	43.8	40.5	25.3

TABLE V
ABLATION STUDY ON THE *SZU_CAR* DATASET

Methods	SSCE	DBFR		CDSF	mAP	mAP_{50}
		3DSBE	SSBR			
Baseline	×	×	×	×	55.6	80.7
SSCE	✓	×	×	×	58.5	84.1
DBFR	×	✓	✓	×	54.8	83.2
CDSF	×	×	×	✓	56.6	83.3
w/o SSCE	×	✓	✓	✓	58.1	84.4
w/o DBFR	✓	×	×	✓	60.1	86.2
w/o 3DSBE	✓	×	✓	✓	58.9	83.2
w/o SSBR	✓	✓	×	✓	59.6	83.7
w/o CDSF	✓	✓	✓	×	60.1	85.6
DAHNet (Full)	✓	✓	✓	✓	**61.5** (+5.9)	**87.0** (+6.3)

higher than the representative method S2ADet. Compared with RTMDet and YOLOv5, DAHNet improves by 7.7% and 8.2%, respectively, demonstrating its strong capability in hyperspectral feature modeling and dense object detection. DAHNet also reaches an mAP50 of 91.2%, exceeding YOLOv8 by 5.0% and YOLOv11 by 6.4%, indicating a more stable balance between precision and recall. In terms of computational complexity and inference efficiency, DAHNet records 149.0 GFLOPs and achieves 27.4 FPS, delivering significant accuracy gains while maintaining reasonable computational cost. Although YOLOv8 attains a higher inference speed of 287.3 FPS, its mAP remains 52.8%, which is notably lower than that of DAHNet. RTMDet and RTDETR exhibit lower computational complexity but both have mAP values below 53%, reflecting limited capacity in capturing detailed spectral–spatial cues. SpecDETR shows considerably higher computational cost of 978.0 GFLOPs and very low inference speed of 3.5 FPS on HOD3K, while achieving an mAP of only 33.2%, which further demonstrates that directly applying DETR-like token-based decoding is inefficient for large-scale HOD. In contrast, DAHNet maintains stable detection performance under diverse spectral distributions and varying object scales, achieving a favorable tradeoff between detection accuracy and inference efficiency. These results further confirm the adaptability and generalization capability of the proposed spectral–spatial hybrid architecture for HOD tasks.

Table IV presents the detection results on the M2SODAI dataset. DAHNet achieves the best overall performance across all metrics, obtaining an mAP of 27.2% and an mAP50 of 64.0%. Compared with YOLOv5 and RTMDet, which achieve mAPs of 23.5% and 20.8%, DAHNet improves by 3.7% and 6.4%, respectively. Relative to the internal baseline with an mAP of 23.4%, DAHNet gains 3.8%, confirming that the proposed spectral–spatial decomposition and aggregation modules consistently contribute to performance enhancement across different datasets. In particular, for small targets, DAHNet achieves an AP_S of 26.7%, outperforming YOLOv5 by 3.4%, indicating its capability to strengthen feature representation and improve the recognition of small-scale objects under low signal-to-noise conditions. Regarding computational complexity and inference efficiency, DAHNet requires 40.5 GFLOPs and runs at 25.3 FPS, achieving significant accuracy improvements while maintaining practical efficiency. Although YOLOv8 and YOLOv11 reach higher frame rates of 143.7 and 137.4 FPS, their mAPs remain 25.7% and 24.9%, both lower than DAHNet. These results demonstrate that the proposed spectral–spatial hybrid structure effectively captures fine-grained spectral features and enables more accurate localization of small maritime targets. DAHNet maintains robust performance under conditions with varying target scales, spectral reflectance differences, and diverse background distributions, further confirming its robustness and generalization ability for HOD tasks.

To further visually validate the effectiveness of our method, we present several detection results and compare them with other methods. As shown in Fig. 5, our method demonstrates significant advantages in both detection accuracy and target localization. Compared to DDQ and DINO, our approach more accurately captures target objects and exhibits superior detection performance in scenarios with strong background interference and occlusion. Particularly in handling occluded objects, our method accurately identifies targets, whereas other methods often result in false positives or missed detections. This advantage highlights the robustness and adaptability of

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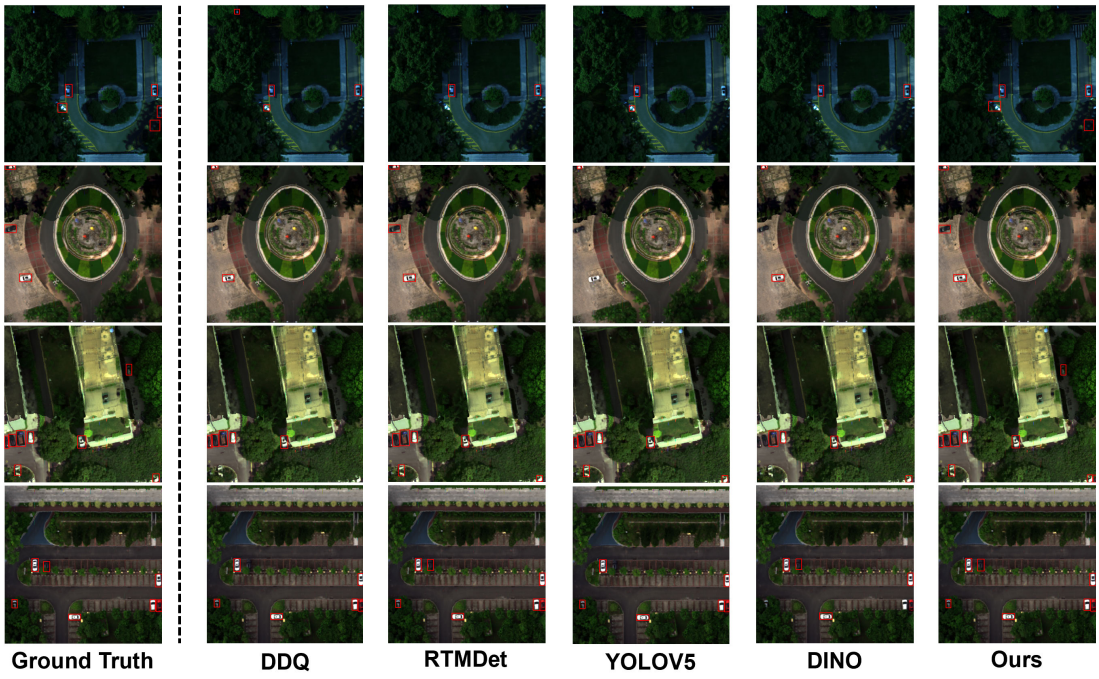


Fig. 5. Comparison of detection results on the *SZU_CAR* dataset using our method, DDQ, RTMDet, YOLOv5, and DINO methods.



Fig. 6. Comparison of detection results on the HOD3K dataset using our method and other comparative methods.

our approach. Furthermore, to assess the performance of our method on small object detection in the M2SODAI dataset, we conducted a visualization analysis. As shown in Fig. 6, our method provides more precise detection on the HOD3K dataset, particularly under varying lighting conditions, where our approach accurately identifies targets even when shadows and illumination changes occur. As depicted in Fig. 7, our method is compared with the DoubleFPN approach presented in the original article. The results indicate that our method excels in detecting small targets, outperforming DoubleFPN by accurately identifying these smaller objects, which often pose significant challenges in complex maritime scenarios.

D. Ablation Study

To evaluate the contribution of each component and substructure within DAHNet, a comprehensive ablation study was conducted on the *SZU_CAR* dataset, and the results are summarized in Table V. Representative structural configurations of several variants are illustrated in Fig. 8, while the remaining settings follow the same design principles. The baseline model consists of two parallel CSPDarknet53 [57] backbones, where feature fusion is performed through elementwise ADD.

Based on this architecture, both additive and subtractive experiments were designed to analyze the effect of each component. In the additive experiments, the SSCE, DBFR, and CDSF modules were individually added to the baseline to measure their independent gains. In the subtractive experiments, the complete DAHNet was used as the reference, and one module or submodule including 3DSBE and SSBR was removed in each case to examine its marginal influence on the overall performance. The SSCE module compresses redundant hyperspectral information and enhances spectral–spatial representations, the DBFR module composed of 3DSBE and SSBR refines spectral and spatial features in parallel, and the CDSF module enables cross-domain interaction and adaptive multiscale fusion. The combination of these additive and subtractive

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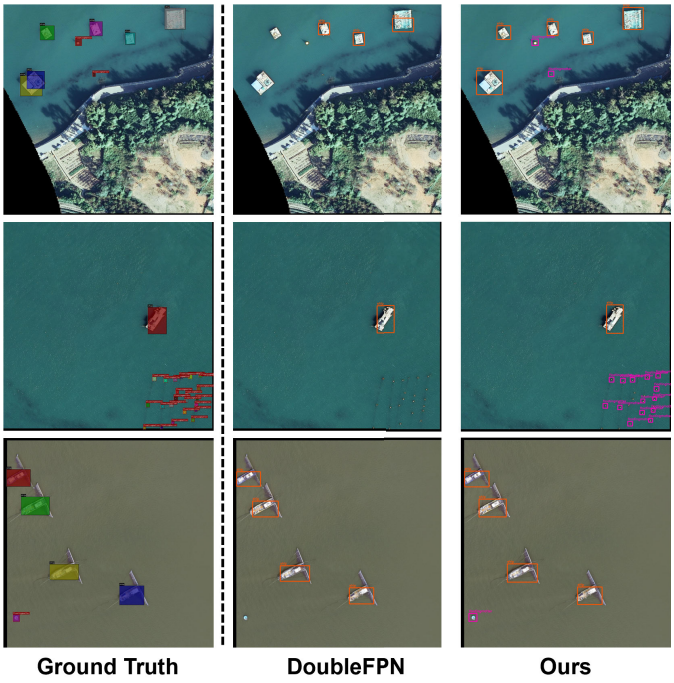


Fig. 7. Visualization comparison on the M2SODAI dataset. Our method demonstrates superior performance in detecting small targets compared to DoubleFPN. The GT bounding boxes are displayed in multiple colors, where each target instance is randomly assigned a distinct color to improve visual clarity in densely distributed regions. The colors do not indicate category information.

TABLE VI

COMPARISON OF DIFFERENT HYBRID ATTENTION MECHANISMS IN THE SSCE STAGE ON THE *SZU_CAR* DATASET

SSCE Variant	**mAP**	**mAP$_{50}$**
Baseline (No Attention)	55.6	80.7
SE	56.1	83.8
CBAM	55.3	80.5
SSCE (Ours)	**58.5**	**84.1**

TABLE VII

COMPARISON OF DIFFERENT FEATURE FUSION MODULES IN THE CDSF STAGE ON THE *SZU_CAR* DATASET

CDSF Variant	**mAP**	**mAP$_{50}$**
Baseline (No Fusion)	55.6	80.7
ViT-based Fusion	54.0	79.4
Swin-based Fusion	53.8	81.0
CDSF (Ours)	**56.6**	**83.3**

analyses provides a comprehensive understanding of how each module and submodule contributes to the detection accuracy and robustness of DAHNet.

1) Effectiveness of the SSCE Module: As shown in Fig. 8(d), removing the SSCE module allows the raw hyperspectral image to be directly fed into the dual-branch structure without any preliminary enhancement. This modification results in a clear decline in detection performance, with mAP$_{50}$ decreasing from 87.0% to 84.4% and mAP decreasing from 61.5% to 58.1%. These results indicate that the SSCE module plays a crucial role in maintaining discriminative spectral–spatial representations. To further validate the design of SSCE, we additionally introduced two hybrid attention mechanisms, SE [47] and CBAM [46], into the same stage for comparison, as shown in Table VI. The SE-based variant demonstrates a slight improvement over the baseline, achieving 56.1% mAP and 83.8% mAP$_{50}$, suggesting that channel reweighting contributes to hyperspectral modeling. However, the CBAM-based variant yields decreased performance at 55.3% mAP and 80.5% mAP$_{50}$, indicating that spatial attention alone is insufficient to handle spectral redundancy and cross-band dependence. The proposed SSCE achieves the best results, reaching 58.5% mAP and 84.1% mAP$_{50}$, which highlights the advantage of its spectral compression and spatial enhancement strategy. These comparisons demonstrate that existing hybrid attention modules, originally designed for RGB imagery, cannot fully exploit the spectral characteristics of hyperspectral data. In contrast, SSCE effectively suppresses redundant spectral responses and enhances salient spatial cues at an early stage, producing more compact and informative representations. This design further strengthens spectral–spatial complementarity, providing a solid foundation for the subsequent refinement and fusion processes within DAHNet.

2) Effectiveness of the DBFR Module: As shown in Fig. 8(c), the DBFR module consists of two refinement branches, namely, the spectral refinement branch (3DSBE) and the spatial refinement branch (SSBR), which are designed to model spectral and spatial information separately during feature extraction. The additive experiment shows that when only the DBFR module is introduced into the baseline model, the performance does not improve, with mAP of 54.8% and mAP50 of 83.2% compared to the baseline values of 55.6% and 80.7%. This indicates that without the early spectral–spatial enhancement and the subsequent cross-domain fusion, DBFR alone cannot fully realize its refinement capability. In contrast, removing the DBFR module from the complete DAHNet leads to a performance drop, where mAP50 decreases from 87.0% to 86.2%, and mAP decreases from 61.5% to 60.1%. These results demonstrate that DBFR plays a vital role in maintaining spectral–spatial feature consistency. By decoupling the learning of spectral and spatial representations, DBFR effectively reduces cross-domain interference and enhances the discriminability of hyperspectral features. Further submodule ablation experiments show that removing 3DSBE reduces mAP from 61.5% to 58.9% and mAP50 from 87.0% to 83.2%, while removing SSBR reduces mAP to 59.6% and mAP50 to 83.7%. Both decreases are larger than those caused by removing the entire DBFR (mAP 60.1% and mAP50 86.2%), indicating that the absence of either branch weakens the overall performance. The 3DSBE primarily strengthens spectral correlation modeling, whereas SSBR focuses on multiscale spatial structure refinement. The collaboration between these two branches forms a balanced refinement mechanism, allowing DBFR to preserve cross-domain feature consistency and significantly improve detection stability and robustness in subsequent fusion stages.

3) Effectiveness of the CDSF Module: As shown in Fig. 8(b), the CDSF module serves as a key component in DAHNet for cross-domain feature fusion. It employs a

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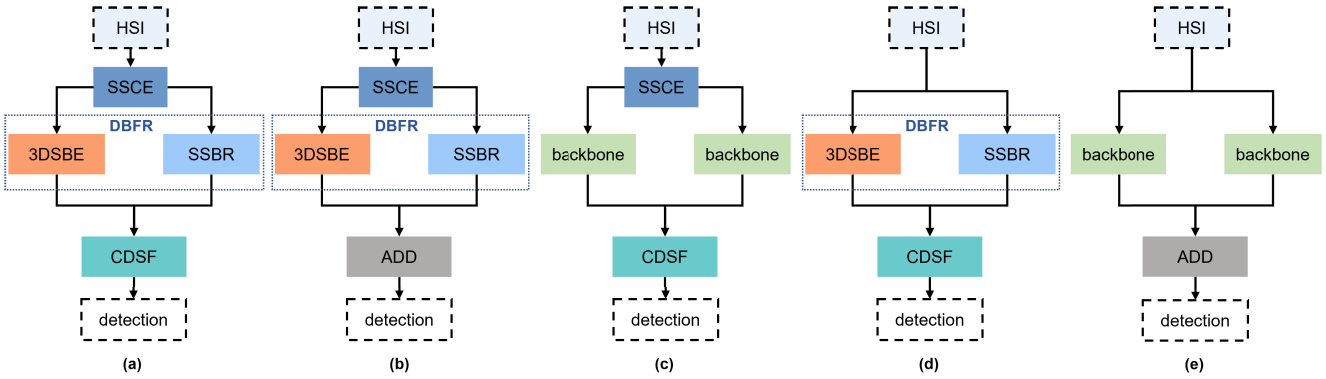


Fig. 8. Structural configurations used in the ablation study. (a) Full DAHNet model integrating all submodules, including SSCE, DBFR (3DSBE + SSBR), and CDSF. (b) Variant replacing the CDSF module with simple elementwise addition (ADD) for feature fusion. (c) Variant removing both DBFR subbranches, where the default Darknet53 backbones directly extract features without spectral–spatial refinement. (d) Variant removing the SSCE module, where the raw HSI is directly fed into the dual-branch structure. (e) Baseline model composed of two parallel Darknet53 backbones with elementwise ADD for fusion and without any SSCE, DBFR, or CDSF module.

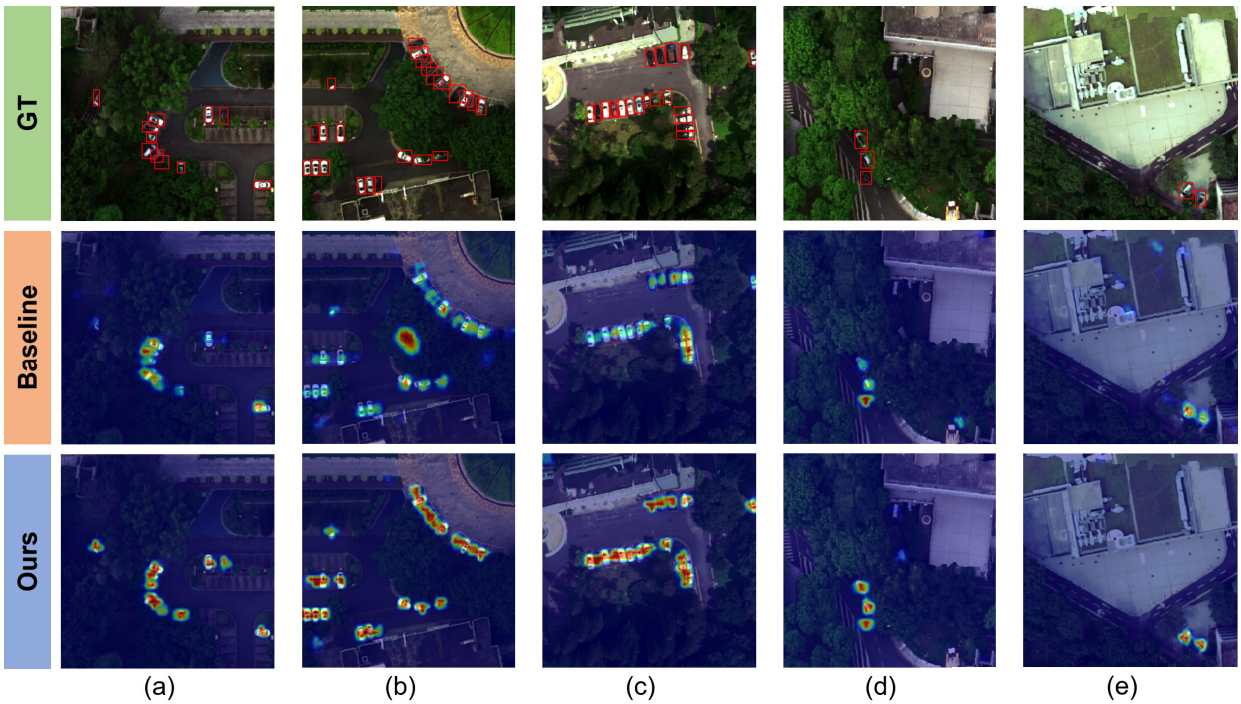


Fig. 9. Heatmap of detection results on the *SZU_CAR* dataset, illustrating the confidence distribution of different methods in detecting targets in the presence of background-induced interference. The intensity of the colors represents the detection confidence levels. (a)–(e) Five representative example scenes selected from the SZU_CAR dataset. Each column corresponds to one example image. The first row shows the ground-truth annotations (GT), the second row presents the detection heatmaps produced by the baseline model, and the third row shows the detection heatmaps generated by the proposed method (DAHNet).

bidirectional cross-attention mechanism to establish dynamic interactions between spectral and spatial representations, enabling semantic alignment and adaptive fusion across domains. When this module is removed, feature fusion is replaced by simple elementwise ADD, resulting in a noticeable decline in performance, with mAP50 decreasing from 87.0% to 85.6% and mAP decreasing from 61.5% to 60.1%. These results demonstrate that CDSF effectively enhances multiscale feature interaction and semantic consistency between spectral and spatial branches. Specifically, spectral features are guided by spatial structures to produce more discriminative responses, while spatial features are refined under spectral constraints that improve boundary precision and intraclass consistency. The additive experiment that introduces only the CDSF module (mAP of 56.6% and mAP50 of 83.3%) shows a limited improvement over the baseline, indicating that CDSF achieves its full potential when collaborating with the preceding enhancement and refinement modules. To further examine whether the performance gains arise from generic attention mechanisms or from the specific design of CDSF, we additionally replaced CDSF with representative Transformer-based fusion modules, including Vision Transformer-based [58] and Swin Transfomer-based [59] designs. As reported in Table VII, these Transformer-style fusion modules do not provide consistent improvements and even lead to performance degradation compared with the

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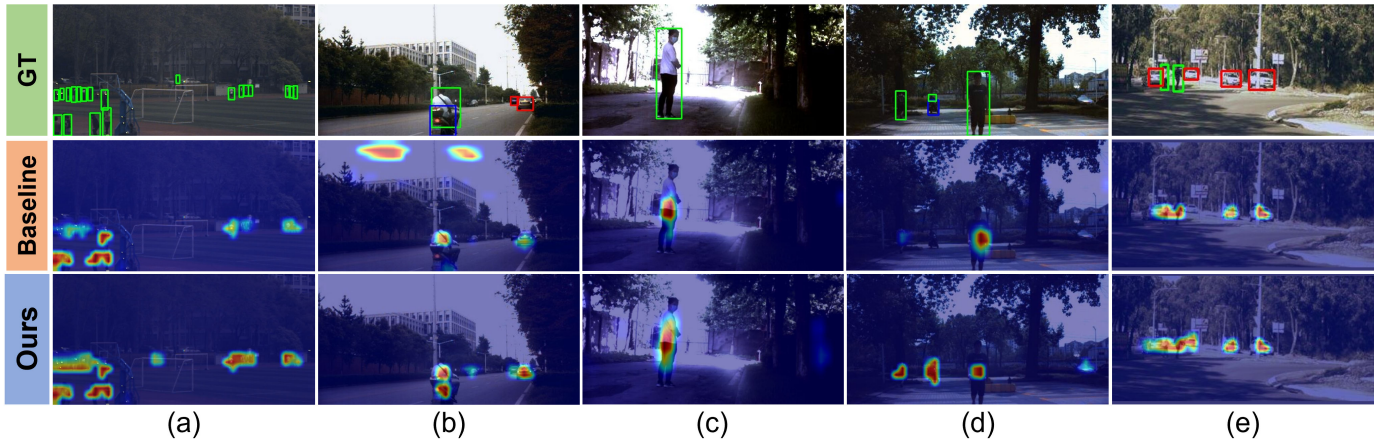


Fig. 10. Heatmap of detection results on the HOD3K dataset, comparing the confidence distributions of our method and other methods in urban environments. Darker colors indicate better detection performance. (a)–(e) Five representative example scenes selected from the HOD3K dataset. Each column corresponds to one example image. The first row shows the ground-truth annotations (GT), the second row presents the detection heatmaps produced by the baseline model, and the third row shows the detection heatmaps generated by the proposed method (DAHNet).

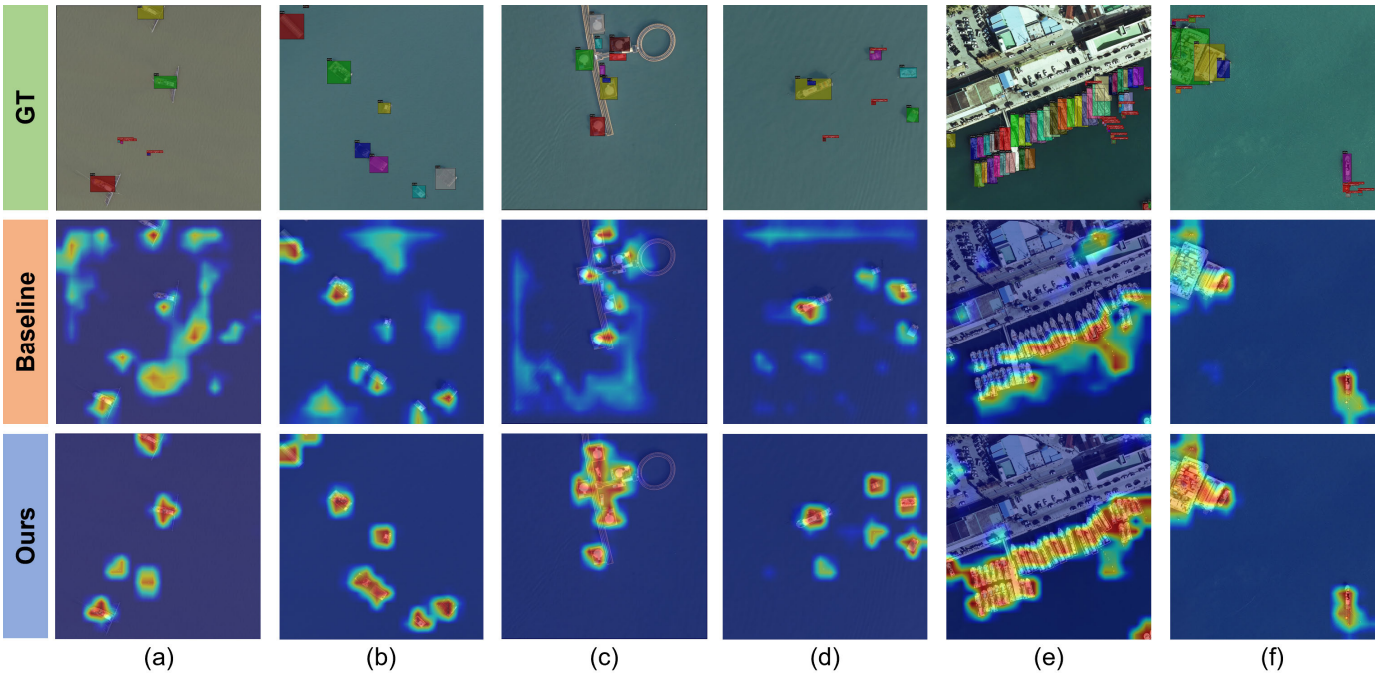


Fig. 11. Heatmap of detection results on the M2SODAI dataset, illustrating the spatial distribution of detection confidence in maritime scenes. The heatmap color gradient represents detection confidence, where warmer colors indicate higher confidence and cooler colors denote lower confidence. The GT bounding boxes are drawn with randomly assigned colors to distinguish densely distributed targets, and these colors do not carry any semantic or categorical meaning. (a)–(f) Six representative example scenes selected from the M2SODAI dataset. Each column corresponds to one example image. The first row shows the ground-truth annotations (GT), the second row presents the detection heatmaps produced by the baseline model, and the third row shows the detection heatmaps generated by the proposed method (DAHNet).

no-fusion baseline. This observation indicates that directly introducing generic self-attention is insufficient for effective spectral–spatial fusion, and highlights the importance of the task-oriented inductive bias embedded in CDSF. Overall, CDSF enables deep semantic integration between spectral and spatial domains, providing crucial support for robust HOD under background interference.

E. Visualization Results

To intuitively demonstrate the effectiveness of the proposed method, we generate both heatmaps and precision–recall (PR) curves on the *SZU_CAR*, HOD3K, and M2SODAI datasets to visualize the model's feature attention and detection stability in different scenarios. The heatmaps are extracted from intermediate feature layers, clearly reflecting the model's focus on target regions within the input hyperspectral images. In the heatmaps, redder areas indicate higher attention, while bluer areas represent lower attention. Each set of heatmaps consists of three rows: the first row visualizes the annotated bounding boxes, the second row displays the heatmap of the baseline model, and the third row presents the heatmap of the proposed method.

On the *SZU_CAR* dataset (Fig. 9), vehicle targets are mainly distributed along complex urban roads, with partial

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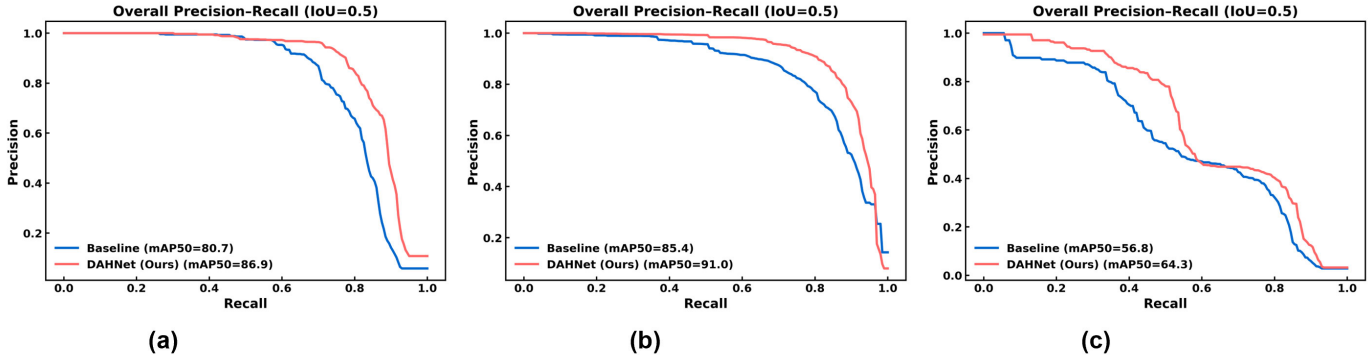


Fig. 12. Overall PR curves at an IoU threshold of 0.5 on three representative HOD datasets. (a) *SZU_CAR*: DAHNet achieves higher precision across a wide recall range, indicating more stable detection confidence. (b) HOD3K: The proposed model maintains superior recall and precision balance under complex illumination conditions. (c) M2SODAI: DAHNet outperforms the baseline by a clear margin, demonstrating strong generalization ability on large-scale aerial scenes.

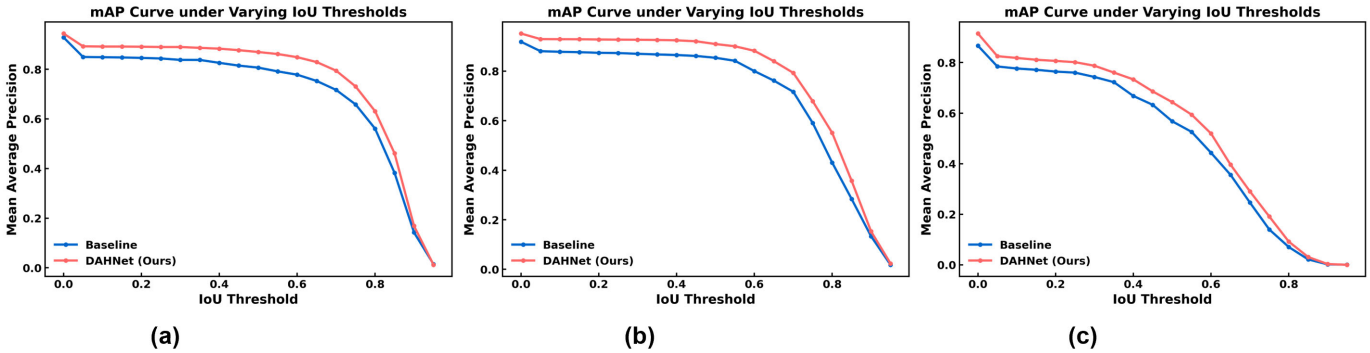


Fig. 13. mAP curves under varying IoU thresholds on three representative HOD datasets. (a) *SZU_CAR*: DAHNet consistently outperforms the baseline as the IoU threshold increases, indicating improved localization precision under background-induced interference. (b) HOD3K: The proposed method maintains a more stable mAP trend across different IoU levels, reflecting robust localization across diverse object categories. (c) M2SODAI: DAHNet shows a clearer advantage at higher IoU thresholds, highlighting its effectiveness in the precise localization of small maritime targets.

occlusion from buildings and trees. The baseline heatmap exhibits dispersed activations and weak focus in occluded regions, while DAHNet produces more concentrated and precise activations that accurately align with vehicle positions. Even under heavy occlusion, the proposed method maintains stable localization, indicating higher robustness in urban environments.

For the HOD3K dataset (Fig. 10), the targets include humans, vehicles, and bicycles distributed across natural campus and road scenes with dense arrangements. The baseline model fails to capture small or crowded objects, leading to scattered attention maps. In contrast, DAHNet's heatmaps show sharper boundaries and stronger responses in dense regions, effectively separating targets from background clutter.

In the M2SODAI dataset (Fig. 11), maritime targets such as ships and buoys appear under low-contrast, low-illumination conditions. The baseline produces broad and weak activations that often merge with the background, whereas DAHNet focuses precisely on the target areas with clear boundaries and high activation strength, highlighting its robustness against challenging maritime scenes.

The PR curves shown in Fig. 12 further validate these observations. Across all datasets, DAHNet consistently maintains higher precision at the same recall level compared with the baseline, demonstrating improved detection confidence and better recall–precision tradeoffs. This improvement reflects the model's enhanced ability to distinguish targets from background noise through the joint spectral–spatial feature learning and adaptive fusion mechanisms. To further analyze localization sensitivity, we additionally report mAP curves under varying IoU thresholds in Fig. 13. Across all three datasets, DAHNet consistently outperforms the baseline over a wide range of IoU values. Notably, the performance gap becomes more pronounced as the IoU threshold increases, indicating that the proposed method improves localization precision rather than merely enhancing classification confidence. This trend is particularly evident on the *SZU_CAR* and M2SODAI datasets, where accurate localization under strict IoU criteria is critical due to background interference and small target sizes. These results further confirm the robustness of DAHNet in precise object localization.

In summary, compared with the baseline, DAHNet exhibits stronger attention concentration, more stable confidence distribution, and higher PR consistency. These results confirm that the proposed spectral–spatial hybrid architecture not only enhances feature expressiveness and small-object recognition but also ensures robust generalization under complex and diverse environmental conditions.

V. Conclusion

In this article, we introduced a novel HOD network, DAHNet, designed to enhance detection performance amid

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background interference by effectively integrating spectral and spatial features. DAHNet employs the SSCE, 3DSBE, SSBR, and the CDSF modules to achieve comprehensive feature extraction and fusion. Through extensive experiments on multiple datasets, including *SZU_CAR*, HOD3K, and M2SODAI, DAHNet consistently demonstrated superior performance across various metrics, particularly in challenging scenarios with complex backgrounds. While DAHNet has shown outstanding results in HOD, there remain challenges in certain specific scenarios. Future research could focus on further optimizing the network architecture or exploring additional techniques tailored for HOD in complex environments to further enhance the model's robustness and detection accuracy.

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